LOST AND FOUND PORTAL

Micro Project Report

Submitted to the APJ Abdul Kalam University, Kerala
for the partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology
in
Computer Science and Engineering

By

KENSTON IGNATIOUS (PRN25CS060)
SREEHARI G (PRN25CS123)
MUHAMMAD SHAROOQU S (PRN25CS
SREENANDAN S (PRN25CS125)

Guided by
Prof. ANISH A AZIZ

During
OCTOBER 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Accredited by NBA)

COLLEGE OF ENGINEERING PERUMON

CAPE - GOVT. OF KERALA, KOLLAM(D), KERALA

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Introduction

The Lost & Found Management System is a Java-based application developed to manage lost and found items in an organized and efficient manner. In colleges and other institutions, students and staff often lose personal belongings such as identity cards, books, wallets, keys, bags, and electronic devices. Finding these items through traditional methods can be difficult and time-consuming. This project provides a centralized system where information about lost and found items can be recorded, stored, searched, and retrieved easily.

The main objective of this system is to simplify the process of reporting lost items and registering items that have been found. Users can enter details such as the item name, category, description, location, and date of loss or discovery. The system helps identify possible matches between lost and found records, making it easier to locate the rightful owners. It also supports claim verification and status updates, allowing items to be tracked from the time they are reported until they are returned to their owners. By maintaining organized records, the system reduces manual effort and improves the efficiency of the recovery process.

This project is implemented using Java and follows the principles of Object-Oriented Programming (OOP). Concepts such as classes and objects, encapsulation, inheritance, abstraction, and polymorphism can be applied to organize the program into manageable components. Collections such as `ArrayList` can be used to store and manage records, while exception handling helps manage invalid inputs and errors. File handling can also be incorporated to preserve records between program executions. The application uses a simple console-based interface, making it suitable for beginner-level programmers and demonstrating how Java and OOP concepts can be used to solve a practical real-world problem.

Scope of the micro-project

The scope of the Lost & Found Management System is to provide a simple and efficient method for managing lost and found items within a college or institution. The system allows users to report lost belongings, register found items, search for items using relevant details, identify possible matches, and maintain records of claims and returned items. It can store information such as item name, category, description, location, date, and current status, helping users and administrators access the required information easily.

The project focuses on reducing manual record-keeping, improving the organization of item details, and making the recovery process more convenient. It is developed as a console-based Java application using Object-Oriented Programming principles. Collections such as Array List can be used to manage records, while file handling can be included to preserve data between program executions. The current scope is limited to basic item reporting, searching, matching, claim verification, and status management. In the future, the system could be extended with a graphical user interface, database integration, user authentication, email notifications, and online access to support larger institutions.

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Technologies used

Technologies Used:

The following technologies were used in the development of this project:

➢ **Programming Language:** Java — used to implement the application logic and functionalities of the Lost & Found Management System.

➢ **UI Toolkit:** Java Swing — used to develop the graphical user interface (GUI) with components such as JFrame, JPanel, JButton, JLabel, JTextField, and other Swing controls.

➢ **Layout Managers:** Java Swing layout managers — used to arrange and organize the interface components within the frames and panels, ensuring a structured and user-friendly interface.

➢ **Event Handling:** Java event handling — used to perform actions in response to user interactions, such as button clicks. It enables users to navigate between screens and execute the available application functions.

➢ **Graphical User Interface Components:** Swing controls, including buttons, labels, text fields, and panels, were used to accept user input, display information, and provide interaction with the application.

➢ **Development Environment:** A Java-compatible IDE was used to write, compile, debug, and run the application.

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Problem Statement

In colleges and educational institutions, students and staff frequently lose personal belongings such as identity cards, books, wallets, keys, bags, and electronic devices. The process of finding these items is often handled through manual records, notice boards, announcements, or informal communication. These methods can be time-consuming and unreliable, making it difficult to identify lost items, locate their rightful owners, and keep track of items that have been found or returned.

The absence of a centralized system for recording and managing lost and found items can lead to misplaced information, delays in recovery, and difficulties in verifying ownership. Therefore, there is a need for a simple and organized system that allows users to report lost items, register found belongings, search for possible matches, and track the status of claims and returned items.

The Lost & Found Management System aims to address these problems through a console-based Java application that maintains organized records and applies Object-Oriented Programming concepts to improve the efficiency and reliability of the recovery process.
Furthermore, the lack of a proper tracking mechanism makes it difficult to determine whether a reported item is still missing, has been found, or has already been returned to its owner. People who find lost belongings may not know where or how to report them, while the actual owners may struggle to obtain information about their missing items. In some cases, similar-looking items can also create confusion during the identification and recovery process.

Another major challenge is the difficulty of maintaining accurate and up-to-date records when information is stored manually. The proposed Lost & Found Management System addresses these challenges by providing a centralized, organized, and easy-to-use solution for reporting and recovering lost belongings.

Literature Review / Background

Lost and found management is an important requirement in educational institutions, offices, and public places, where people frequently misplace personal belongings. Traditionally, these activities are handled using manual registers, notice boards, announcements, and direct communication. With the advancement of software technology, computerized systems have become a more organized approach to managing information. Digital lost and found applications provide facilities to record item details, search available records, identify possible matches, and update the status of reported items. These features help reduce manual effort and improve the efficiency of the recovery process.

The application provides an interactive graphical user interface (GUI) developed using **Java Swing**. Different frames are designed for specific operations, such as registering lost items, reporting found items, searching for records, managing claims, and viewing item details. Swing components, including labels, text fields, buttons, tables, and other controls, are used to collect user information and display results. Layout managers organize these components within each frame, providing a structured and user-friendly interface. Action listeners are associated with buttons so that users can perform operations such as adding records, searching for items, updating details, and changing item status through simple button clicks.

The project follows Object-Oriented Programming (OOP) principles by organizing the application into classes and objects that represent real-world entities such as users, items, and claims. Encapsulation helps protect data, inheritance supports the reuse of common properties and methods, abstraction defines essential behavior, and polymorphism allows different classes to implement common operations in different ways. Event handling in Java Swing connects user interactions with the program's functionality, while exception handling helps manage invalid inputs and unexpected errors. Collections and file handling can also be used to manage and preserve records, depending on the implementation. Overall, the project combines Java programming, Swing GUI development, and OOP concepts to provide an organized and convenient solution for managing lost and found belongings.

Implementation

Tools and Technologies Used:

- **Java Swing:** Swing is used to create the graphical user interface. Components such as JFrame, JPanel, JLabel, JTextField, JButton, JTable, and JOptionPane are used to design application windows, collect user input, display records, and show messages

- **Layout Managers:** Layout managers are used to arrange interface components within frames and panels. They help maintain a structured and organized appearance across different screens.

- **Event Handling:** Action listeners are attached to buttons to execute specific operations when users interact with the application. These actions include adding records, searching for items, updating information, and managing item status.

- **Object-Oriented Programming:** OOP concepts such as classes and objects, encapsulation, inheritance, abstraction, and polymorphism are used to organize the program into manageable and reusable components.

- **Development Environment (IDE):** A Java-compatible IDE, such as Eclipse or VS Code can be used to write, compile, debug, and execute the application. The specific IDE depends on the development environment used by the project team.

The implementation begins with designing the application frames and arranging their components using suitable layout managers. Each frame is connected to the required functionality through button actions and event listeners. Input fields are used to collect item details, while tables or other display components present the stored information. Validation and exception handling are incorporated to manage missing fields, invalid inputs, and incorrect item identifiers. The application's classes and methods handle the underlying operations and maintain consistency between records. If file handling or database storage is implemented, it can be used to preserve records beyond a single program execution.

Overall, the system combines Java, Swing components, layout managers, event handling, and OOP principles to create an interactive application that is more organized and convenien

Details of Java classes and architecture

The Lost & Found Management System is developed as a Java desktop application using Java Swing for the graphical user interface.

The Lost & Found Management System is developed using Java Swing to create a graphical user interface (GUI). The application consists of multiple Java classes, with each class representing a specific screen or function of the system.

The main classes include:

- **WelcomeFrame.java:** Provides the welcome screen and navigation options.
- **LoginFrame.java:** Provides the login interface for users.
- **RegisterFrame.java:** Allows new users to access the registration interface.
- **HomeFrame.java:** Displays the main dashboard and navigation options.
- **ReportLostFrame.java:** Provides a form for reporting lost items.
- **ReportFoundFrame.java:** Provides a form for registering found items.
- **MyReportsFrame.java:** Displays the interface for viewing user reports.
- **SearchFrame.java:** Provides an interface for searching lost and found items.
- **ItemDetailsFrame.java:** Displays the details of an item.
- **ClaimFrame.java:** Provides the interface for claiming an item.
- **AdminFrame.java:** Provides the administrative dashboard.
- **ManageItemsFrame.java:** Provides an interface for managing item records.
- **ClaimReviewFrame.java:** Provides an interface for reviewing item claims.
- **ReportsFrame.java:** Provides an interface for viewing reports.

  Each class extends JFrame to create an application window. Java Swing components such as JButton, JLabel, JTextField, and JPanel are used to design the interface.

2. System Architecture

The application follows a modular, GUI-based architecture in which each Java class represents a separate screen. This approach makes the project easier to organize, maintain, and extend.

The system is divided into three conceptual layers:

- **Presentation Layer:** Displays the application screens, forms, labels, text fields, and buttons using Java Swing.
- **Application Logic Layer:** Intended to handle operations such as user authentication, item searching, reporting, and claim management in future development.
- **Data Layer:** Intended to store and retrieve user information, lost and found item records, and claim details through a database in a future development phase.

  Currently, the implementation focuses on the presentation layer. The application logic and database integration have not yet been implemented.

3. Key Modules and Code Overview

GUI Creation: Each screen is created using a Java class that extends JFrame. Methods such as setTitle(), setSize(), and setVisible() configure and display the application window.

User Interface Components: Components such as JButton, JLabel, and JTextField are used to create buttons, display text, and accept user input.

Example:

```
JButton submitButton = new JButton("Submit");
JButton clearButton = new JButton("Clear");
JButton backButton = new JButton("Back");
```

These buttons are currently interface elements and do not yet perform their intended operations.

Database Connection and CRUD Operations: Database connectivity and CRUD operations (Create, Read, Update, and Delete) are not part of the current implementation. They may be added in a future phase to support data storage, item searching, record management, and claim processing.

4. Development Status

The current development phase focuses on designing the required application screens and placing the necessary interface components. Future development will include button functionality, user authentication, database connectivity, item management, and claim processing.

5. Overall Architecture Diagram

The application's planned architecture can be represented as follows:

```
User
↓
Java Swing User Interface
↓
Individual JFrame Classes
↓
Navigation Between Screens
↓
Application Logic (Future Development)
↓
Database / Data Storage
```

6. Code Structure and Design Approach

```
import javax.swing.*;
public class ExampleFrame extends JFrame {
   public ExampleFrame() {
      setTitle("Example Screen");
      setSize(600, 400);
      setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);
      JButton exampleButton = new JButton("Example");
      add(exampleButton);
      setVisible(true);
   }
   public static void main(String[] args) {
      SwingUtilities.invokeLater(() -> new ExampleFrame());
   }
}
```

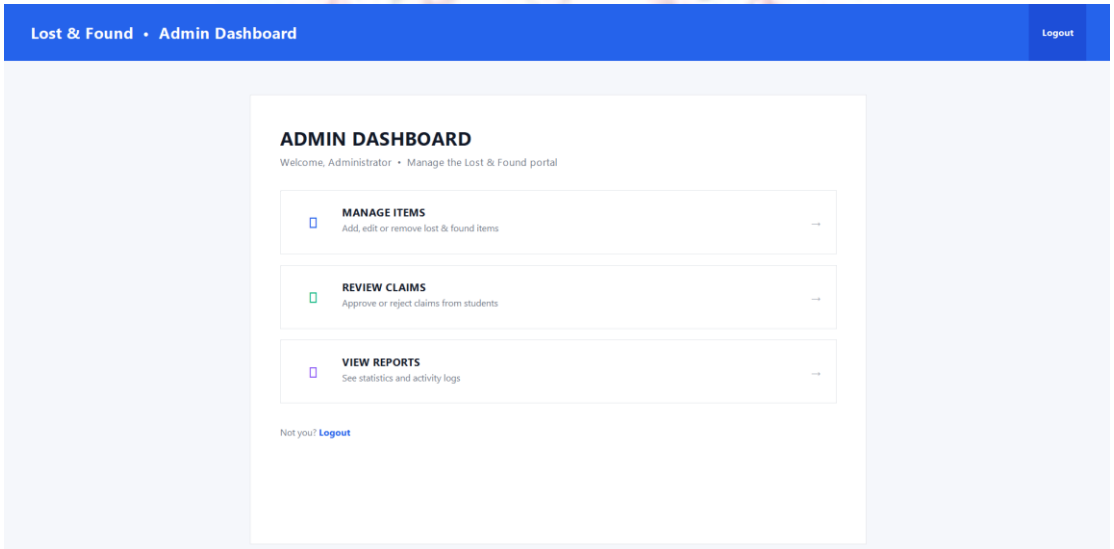
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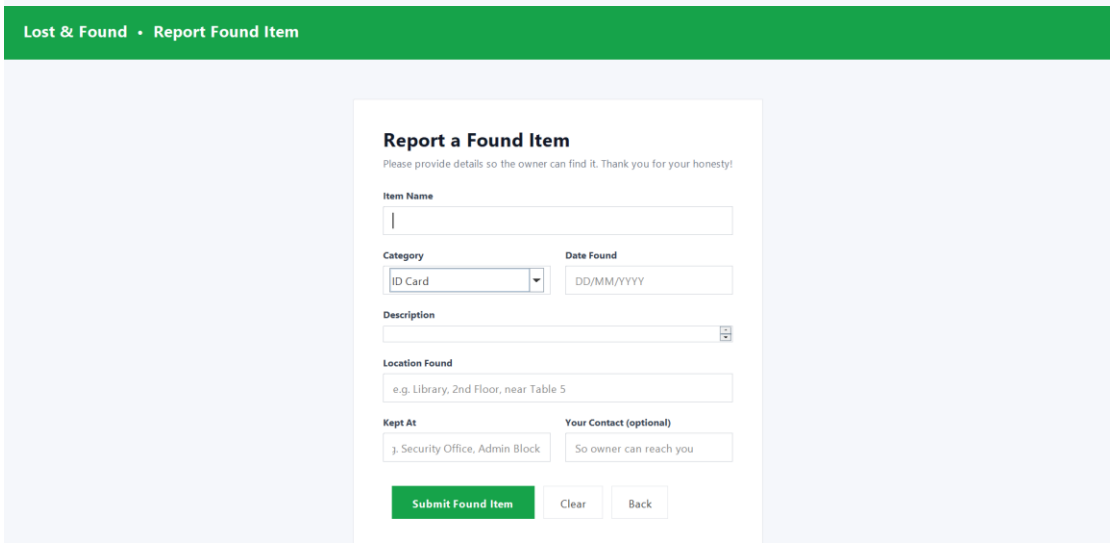
Results & Discussion

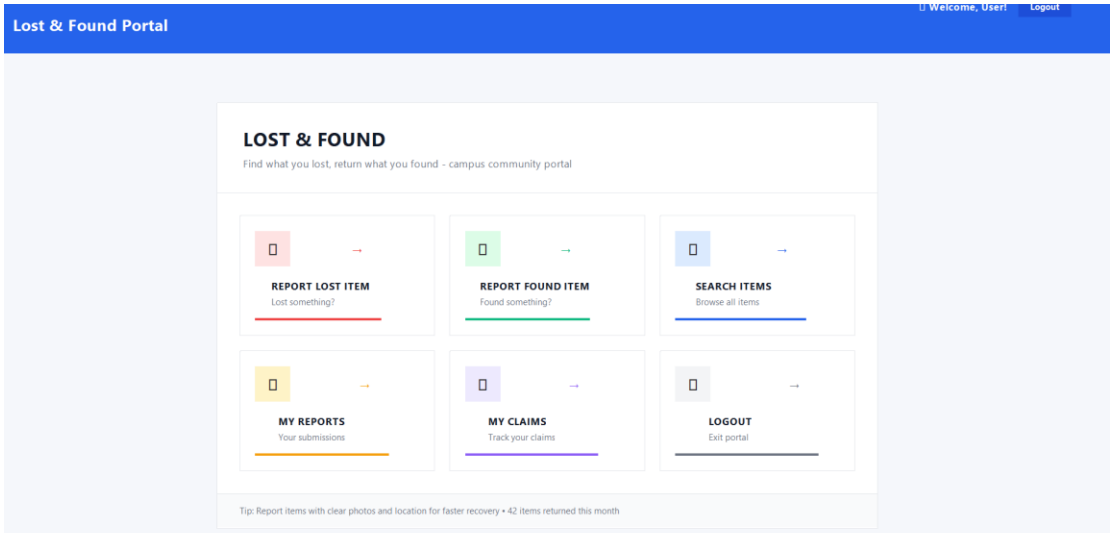
The current version of the Lost & Found Management System focuses on developing the graphical user interface using Java Swing. It consists of fourteen main JFrame classes representing the major screens of the application.

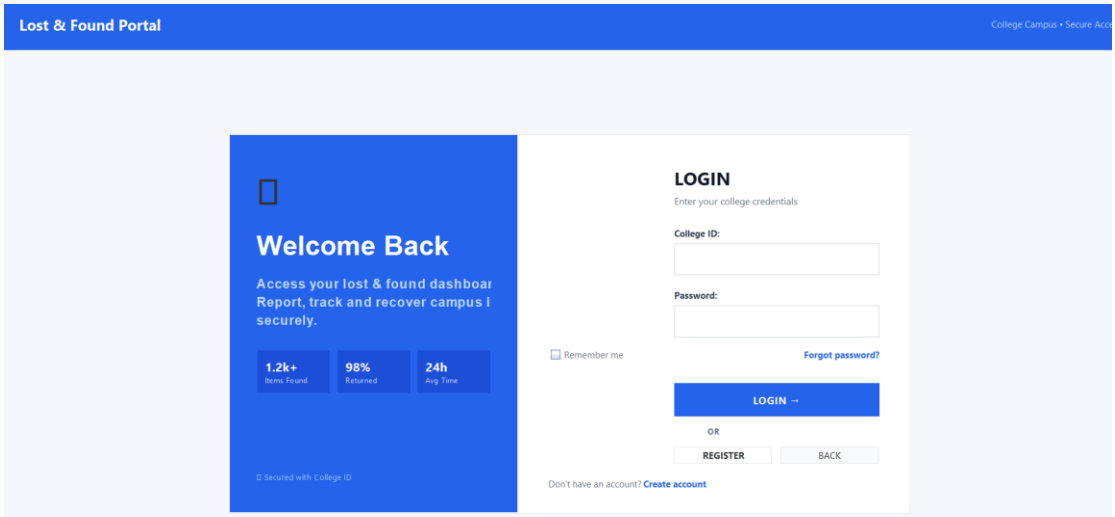
The modular structure separates the different user-facing features into individual Java classes. Buttons and navigation elements provide the structure for interaction, with some navigation functionality implemented and other actions reserved for future development.

The current architecture provides a foundation for extending the application with input validation, authentication, database connectivity, CRUD operations, item matching, and claim processing in subsequent development stages.

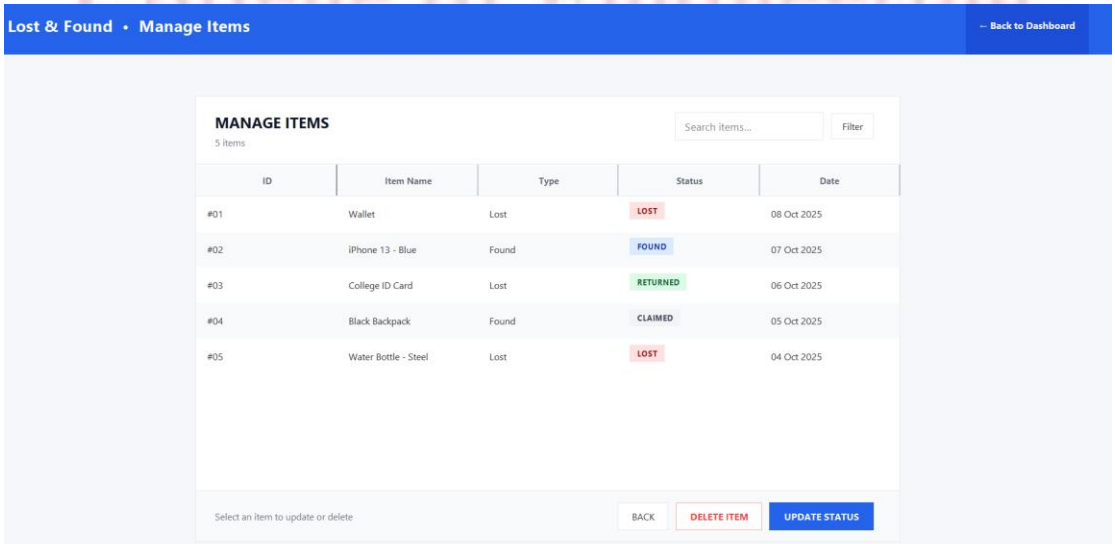








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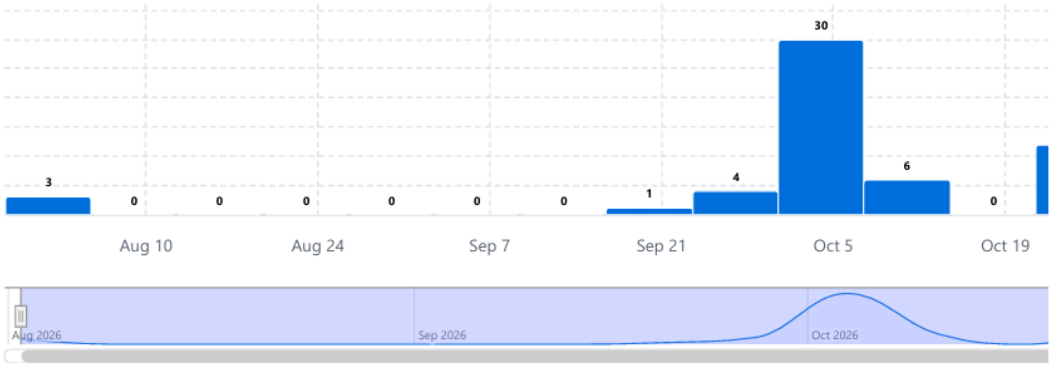
Conclusion

The Lost & Found Management System is a Java-based application developed using Java Swing to simplify the process of reporting and finding lost items within a college campus. It provides a user-friendly graphical interface and demonstrates the practical use of Java, Object-Oriented Programming, Swing components, layout managers, and event handling. This micro project helped us gain practical experience in developing GUI-based applications to solve real-world problems. In the future, the system can be enhanced with database connectivity, user authentication, and advanced item-matching features.



Commits over time

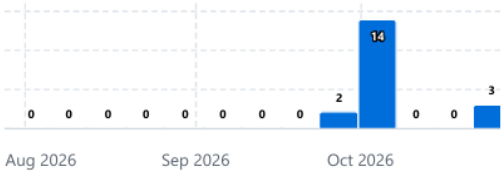
Weekly from Aug 1, 2026 to Oct 25, 2026



sreenandan2007

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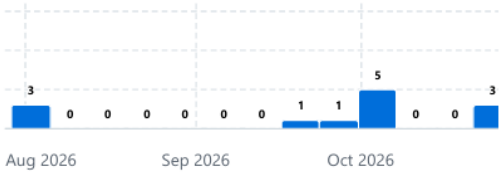
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kenston07

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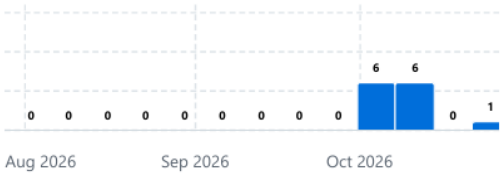
13 commits 2,320 ++ 1,072 --



msharooqu27

#3

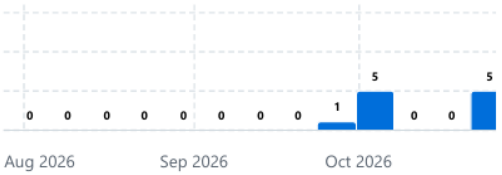
13 commits 2,163 ++ 994 --



sreehari0072

#4

11 commits 3,960 ++ 1,903 --



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